

Data visualisation with clcharts: : CHEAT SHEET

Basics

clcharts creates charts in the House of Commons Library style and is based on [ggplot2](#).

ggplot2 is based on the grammar of graphics, the idea that you can build every graph from the same components: a **data** set, a **coordinate** system, and **geoms**—visual marks that represent data points.

Quickstart

LINE CHART

```
plot <- ggplot(
  data = <data>,
  mapping = aes(x = <x>, y = <y>)
) +

geom_line(color = commonslib_color("commons_green"), size = 1.1) +

labs(
  x = <x>,
  y = <y>
) +

scale_x_continuous(breaks = seq(0, 10, 2)) +
scale_y_continuous(limits = c(0, 20)) +

theme_commonslib()

plot <- plot + add_commonslib_titles(
  plot,
  title = <title>,
  subtitle = <title>
)
# Save the plot
save_png(plot, "<plotname>.png", width = 8, height = 5)
```

aes maps <x> to x and <y> to y. This can be set here to apply to all geoms, or set in each geom.

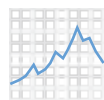
Set x and y axes labels here

Control axes: x is from 0 to 10 in steps of 2, y is between 0 and 20

Common chart types

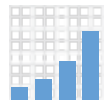
Two variables

plot <- ggplot(data = <data>, mapping = aes(x = <x>, y = <y>)) +



Line chart

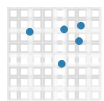
plot + **geom_line**(size=1.1)



Column chart

plot + **geom_col**(width=0.8, fill=<colour>)

Plots bars at discrete <x> with heights <y>

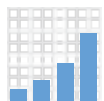


Scatter chart

plot + **geom_point**(shape=16)

1 discrete variable

plot <- ggplot(data = <data>) +



Bar chart

plot + **geom_bar**(mapping = aes(x=<x>, width=0.8, fill=<colour>)

<x> is categorical, geom_bar plots the number in each category

Theme colours

	commons_green		tangerine
	ocean_green		cerulean_blue
	pine_green		pacific_blue
	grape		burnt_orange
	lilac		cherry

Get theme colours by: commonslib_color(<color_name>).

Get political party colours from commonslib_party_color(<party_name>).

Palettes

main	
all	
green	
greenpurple	

Scales map data values to the visual values of an aesthetic. This can be mapping 'color' or 'fill' to one of the available palettes

scale_color_commonslib(palette=<palette_name>)

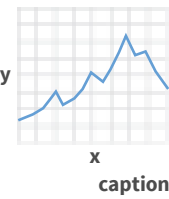
scale_fill_commonslib(palette=<palette_name>)

Saving

```
save_svg(plot, filename, width, height)
save_png(plot, filename, width, height, dpi)
```

width, height in inches.
dpi = dots per inch (default 400)
100-150dpi, lower resolution
300 dpi: standard
600 dpi: high detail

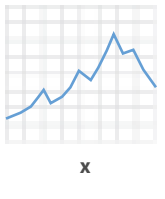
Chart adjustment and annotation



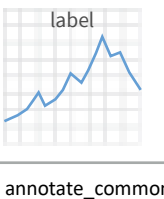
labs(caption=<caption>) +
theme_commonslib(
caption_position=<left|right>
)



labs(x=NULL)



labs(y=NULL)



annotate_commonslib(
x=<x>, y=<y>,
label=<label>,
size=4.2)

Equivalent to 12pt, size is in mm (1 pt = 0.35mm)

Puts annotation at <x>,<y>

coord_flip()
Flips chart

scale_y_continuous(expand=c(0,0))
Turns of axis expansion for y axis